

Lecture 9

Sept 22

## Learning objectives

- Explain why Laplace's "mean of probability" and "mean of error" lead to the posterior median, which divides posterior probability in half and minimizes posterior expected absolute error.
- Describe Laplace's criteria for an error curve and distinguish choosing a model for errors from choosing an estimate of the true value.
- Use conditional probability and the law of total probability to explain Laplace's inverse-probability rule for equally likely causes, and identify the conditional independence assumption in his argument about repeated observations.
- Distinguish joint, conditional, and marginal probabilities, and diagnose Laplace's error.
- Distinguish evaluating a rule across repeated samples with a fixed unknown cause from evaluating an answer using posterior probabilities conditional on the observed data and a specified prior.

Choose your "mean"!

$P(\text{Parameters} | \text{data})$



- Like the sample mean
- ① Equally probable that the true value is before or after my chosen mean
    - mean of probability
    - posterior median
  - ② minimize the sum of errors
    - astronomical mean
    - minimizes posterior expected loss under absolute loss

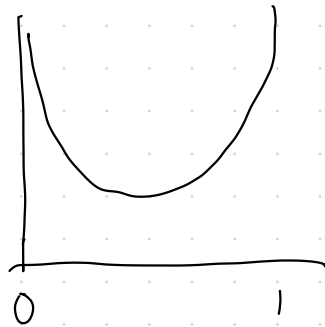
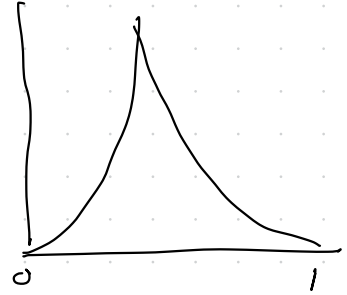
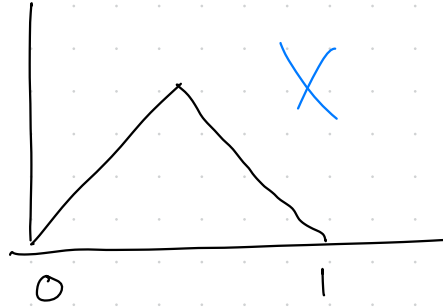
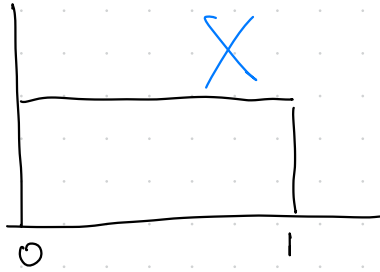
Choose your curve!

① Symmetric about zero

② Decrease as we move away from zero

③ Error should have a prob. distn

# Principle of insufficient reason



Wants: decrease away from truth  
non-constant derivatives

$$\frac{d}{dx} \phi(x) = -\phi(x)$$

$$\phi(x) = \frac{n}{2} \exp(-n|x|)$$

Laplace and conditional probabilities

Recall:  $P(a|b) = \frac{P(a \text{ and } b)}{P(b)}$

conditional      joint      marginal

$$P(a|b) \propto P(a \text{ and } b) \quad \leftarrow$$

$$P(a|b) = C \cdot P(a, b)$$

↑  
constant in a

$$P(a, b) \propto P(a|b)$$
$$P(a, b) \propto P(b|a)$$

He made a huge error.

The curve:  $\phi(x) = \frac{m}{2} \exp(-m|x|)$  (in modern notation)

He needs  $\iint \phi(x|p, g, m) \phi(m|p, g) dm dx$

and he writes  $= \iint \phi(x, p, g|m) \phi(m|p, g) dx dm$

$$\phi(x|p, g, m) \propto \phi(x, p, g|m)$$

$$\phi(x|p, g, m) = \phi(x, p, g|m) / \underbrace{\phi(p, g|m)}_{\text{depends on } m}$$

scale of the error curve

unknown

What to do when you make mistakes?

Unhappy with  $e^{-n|x|}$

- Start over

- Troubleshoot in general

He did everything from first principles, so what's left?

# Handout

Task 1

$$P(A) = \frac{50}{100}$$

$$P(B) = \frac{50}{100}$$

$$P(\text{red} | A) = \frac{45}{50}$$

$$P(\text{red and } A) = \frac{45}{100}$$

## Combining conditional distributions

V3 WORKED VE

Worked answers to v3 | Questions retained; answers and added working are in red.

Each ticket has a source (A or B) and a color (red or blue). Mix all 100 tickets and draw one at random. Every ticket is equally likely; you do not choose a source first.

To normalize a red/blue pair, divide both numbers by their sum. For a source's count pair, that sum is its normalizing constant.

### Task 1 | Equal numbers of tickets from A and B

| Source | Red tickets | Blue tickets | Total |
|--------|-------------|--------------|-------|
| A      | 45          | 5            | 50    |
| B      | 15          | 35           | 50    |
| Total  | 60          | 40           | 100   |

#### 1. Identify the probabilities

Calculate  $P(A)$  and  $P(B)$ . Among tickets from A, what fraction are red? Among all tickets, what fraction are b

Task 1 Q2

$$P(A) \cdot P(\text{red} | A) = \dots \frac{50}{100} \cdot \frac{45}{50}$$

| Source  | Red contribution | Blue contribution |
|---------|------------------|-------------------|
| A       |                  |                   |
| B       |                  |                   |
| Overall |                  |                   |

$\frac{3}{5}$

$\frac{2}{5}$

Q3

$$P(A) \cdot \text{count}(\text{red} | A)$$

| Source   | Weighted red count | Weighted blue count | Total |
|----------|--------------------|---------------------|-------|
| A        |                    |                     |       |
| B        |                    |                     |       |
| Combined |                    |                     |       |

$\frac{3}{5}$

$\frac{2}{5}$

Task 2 Q1

$$P(\text{red} | A) = \frac{9}{10}$$

$$P(\text{red} | B) = \frac{27}{90}$$

### Task 2 | Unequal numbers of tickets from A and B

The drawing rule is unchanged: select one ticket at random from all 100.

| Source | Red tickets | Blue tickets | Total |
|--------|-------------|--------------|-------|
| A      | 9           | 1            | 10    |
| B      | 27          | 63           | 90    |
| Total  | 36          | 64           | 100   |

1. Identify what changed

Calculate the new  $P(A)$  and  $P(B)$ . Did either source's conditional color distribution change? Calculate and blue probabilities directly from the ticket counts as a reference.

# Task 2 Q2

| Calculation                                | Probability of red                                                                 | Probability of blue                                                                 |
|--------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| From the ticket counts                     |  |  |
| Normalize within each source, then combine |  |  |
| Use count pairs, then normalize at the end |  |  |

# Laplace's error, in modern notation

d: observed data    x: unknown correction    m: unknown error-distribution parameter

## UNNORMALIZED CURVE

$$K_m(x) = f(x, d \mid m)$$

## NORMALIZING CONSTANT

$$Z_m = \int K_m(x) dx = f(d \mid m)$$

## WEIGHT ON m

$$w(m) = f(m \mid d)$$

For a fixed m:  $f(x \mid d, m) = \frac{K_m(x)}{Z_m} \propto_x K_m(x)$  Valid as x varies.

## Correct mixture

$$f(x \mid d) = \int_0^\infty \frac{K_m(x)}{Z_m} w(m) dm$$

drops

$1/Z_m$

## The mistaken substitution

$$\tilde{f}(x) \propto_x \int_0^\infty K_m(x) w(m) dm$$

After normalizing in x,  
the weight on m becomes

$$\tilde{w}(m) = \frac{Z_m w(m)}{\int_0^\infty Z_u w(u) du}$$

not the intended  
weight w(m).

$Z_m$  is constant in x for fixed m, but it changes across the m values being combined.

A final division fixes total area; it does not restore the intended relative weights.

# The ticket exercise: replace the integral by a sum

m is the source (A or B); x is the color (red or blue). Pairs are written (red, blue).

| Symbol   | Meaning in the exercise               | Task 1                           | Task 2                          |
|----------|---------------------------------------|----------------------------------|---------------------------------|
| $K_m(x)$ | Counts for each color within source m | $A : (45, 5) \quad B : (15, 35)$ | $A : (9, 1) \quad B : (27, 63)$ |
| $Z_m$    | Row total (sum over colors)           | $Z_A = 50, \quad Z_B = 50$       | $Z_A = 10, \quad Z_B = 90$      |
| $w_m$    | Probability of source m               | $w_A = 0.5, \quad w_B = 0.5$     | $w_A = 0.1, \quad w_B = 0.9$    |

Normalize within each source, then combine

$$g(x) = \sum_{m \in \{A, B\}} w_m \frac{K_m(x)}{Z_m}$$

Resulting distributions (red, blue)

Use count pairs; normalize at the end

$$h(x) = \frac{\sum_{m \in \{A, B\}} w_m K_m(x)}{\sum_{m \in \{A, B\}} w_m Z_m}$$

|        |              |                                            |
|--------|--------------|--------------------------------------------|
| Task 1 | (0.60, 0.40) | $(30, 20)/50 = (0.60, 0.40)$               |
| Task 2 | (0.36, 0.64) | $(25.2, 56.8)/82 \approx (0.3073, 0.6927)$ |

Task 1 drops the same divisor, 50. Task 2 drops different divisors, 10 and 90.

Metaphysical difference of opinion

Frequentist / Legendre

Bayesian / Laplace

Next class

What's the difference?

Two bags      Mostly blue      (8 blue, 2 red)  
                     Mostly red      (2 blue, 8 red)

Rule A: Follow the ball      (if you see blue, guess blue)  
                                         if you see red, guess red

Rule B: Always guess blue

Frequentist evaluation

What bag is hidden

Mostly blue

Mostly red

Rule A

Rule B

Which one is better?

## Bayesian evaluation

Say you believe there's a 50/50 chance of each bag.

If you see a blue ball and you use Rule A

What about Rule B?

